

Enhancing academic integrity among students in GenAI Era: A holistic framework

Tareq Rasul^a, Sumesh Nair^a, Diane Kalendra^a, M.S. Balaji^{b,*}, Fernando de Oliveira Santini^{c,d}, Wagner Junior Ladeira^{c,d}, Raouf Ahmad Rather^e, Naveed Yasin^{f,j}, Raul V. Rodriguez^d, Panagiotis Kokkalis^g, Md Wahid Murad^h, Md Uzir Hossainⁱ

^a Australian Institute of Business (AIB), Adelaide, Australia

^b Rennes School of Business, France

^c Universidade do Vale do Rio dos Sinos (UNISINOS), Porto Alegre, Brazil

^d School of Business, Woxsen University, Hyderabad, India

^e Scientific Researcher, Kashmir, India

^f Abu Dhabi School of Management, United Arab Emirates

^g Rochester Institute of Technology of Dubai, United Arab Emirates

^h University of South Australia, Australia

ⁱ Putra Business School, Malaysia

^j University of Roehampton London, United Arab Emirates Campus

ARTICLE INFO

Keywords:

Generative AI
Academic integrity
Higher education
Students
Stakeholders

ABSTRACT

The introduction of Artificial Intelligence (AI), specifically Generative AI (GenAI), has significantly transformed the higher education landscape. Despite the opportunities GenAI offers to students, they pose significant challenges for academic integrity. Thus, it is crucial for higher education institutions (HEI) to balance the use of GenAI for enhancing the learning experience of students with its ethical and responsible use in their educational journey. The present study proposes a comprehensive academic integrity framework focusing on three key stakeholders: students, educators, and institutions. We propose eight strategies ranging from collaborative learning for students to developing a comprehensive GenAI policy for institutions in maintaining academic integrity among students in HEI. Furthermore, we identified four challenges, namely financial, strategic, operational, and cultural, in the implementing a comprehensive academic integrity framework in the GenAI era. This study offers significant insights for HEI to maintain academic integrity among students in the GenAI era.

1. Introduction

The introduction of Artificial Intelligence (AI) has significantly transformed the learning and teaching approaches in the higher

* Corresponding author.

E-mail addresses: tfrasul@gmail.com, tareq.rasul@aib.edu.au (T. Rasul), sumesh.nair@aib.edu.au (S. Nair), diane.kalendra@aib.edu.au (D. Kalendra), balaji.makam@rennes-sb.com (M.S. Balaji), fosantini@unisinis.br (F.O. Santini), wjladeira@gmail.com (W.J. Ladeira), r.raouf18@gmail.com (R.A. Rather), Yasinresearch@gmail.com (N. Yasin), vice.president@woxsen.edu.in (R.V. Rodriguez), pxkcd@rit.edu (P. Kokkalis), Wahid.Murad@unisa.edu.au (M.W. Murad), hossainuzir@gmail.com (M.U. Hossain).

<https://doi.org/10.1016/j.ijme.2024.101041>

Received 21 March 2024; Received in revised form 2 August 2024; Accepted 7 August 2024

Available online 14 August 2024

1472-8117/© 2024 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

education context (Crompton & Burke, 2023; Zawacki-Richter et al., 2019). Particularly, the integration of AI tools and applications allowed higher education institutions (HEIs) to offer a personalized learning experience for students, thereby increasing their engagement in the learning process (Wang et al., 2021). The recent emergence of Generative AI (GenAI) has resulted in further transformation of how students learn and acquire knowledge. GenAI refers to computational techniques that generate new and meaningful content in the form of text, images, videos, or audio from training data (Feuerriegel et al., 2024). Specifically, GenAI tools, such as ChatGPT, Copilot, and Gemini, stand out, as they are interactive, collaborative, and responsive and can create content instantly. This provides significant opportunities for students to enhance their learning, develop relevant skills, and develop an in-depth understanding of the subject matter (Yilmaz & Yilmaz, 2023). However, these benefits have come with significant disruptions in student assessments and pedagogical approaches (Fui-Hoon Nah et al., 2023). This has prompted HEIs and educators to explore approaches for the responsible use of the GenAI tools by students in their educational journey (Adigüzel et al., 2023).

The widespread adoption of GenAI tools by students has triggered intense debate in HEIs regarding its impact on student academic integrity and learning outcomes (Abdaljaleel et al., 2024). On the one hand, the capabilities of GenAI tools allow HEIs to personalize learning content, enhance research skills, and support self-learning for students. Furthermore, its interactive interface and real-time responses facilitate immersive learning experiences and help students understand better complex concepts (Lim et al., 2023). On the other hand, various concerns have been raised regarding the use of these tools. The authenticity of student work is questioned because of the ease with which these tools can create content and facilitate plagiarism (Cotton et al., 2024; Farrokhnia et al., 2023). Furthermore, despite knowing that using these tools for assessments constitutes cheating, the majority of US students persist in using GenAI tools for assessments (Lee et al., 2024). Another key challenge is that over-reliance on GenAI tools reduces attention, decreases memory storage, and declines critical thinking skills, which will have a negative impact on students' intellectual development (Chan, 2023). These challenges highlight the need for HEIs to develop a comprehensive approach for the responsible and ethical use of GenAI tools by students in their learning process. Despite recent studies attempting to address this issue (Cotton et al., 2024; Liu et al., 2023; Roe & Perkins, 2023), there is limited research proposing a comprehensive approach for maintaining academic integrity among students in the GenAI era.

Given the potential of students to engage in academic dishonesty using GenAI tools, HEIs should go beyond the traditional approaches for assessing their learning. Previous research has mostly focused on the ability of GenAI tools to complete specific assessments and the distinguishing features of human-written and AI-generated content (Herbold et al., 2023; Newton & Xiromeriti, 2023; Rasul et al., 2023). In response to this, HEIs implemented preventive measures for the use of GenAI tools in their academic programs. For example, several universities within the UK Russell Group have banned the use of GenAI tools for assessment purposes over fears of plagiarism (Wood, 2023). Similarly, universities globally are reviewing their plagiarism policies to address the concerns presented by GenAI tools in assessing student learning (Chan, 2023). As addressing the challenges of these tools pose to academic integrity is a multifaceted in nature, a more comprehensive and inclusive strategy involving key stakeholders, such as, students, educators, and the institution is necessary. By collaborating together, the stakeholders can create an environment that promotes academic honesty in HEIs, where students responsibly use GenAI tools in their learning process.

The present study proposes a comprehensive stakeholder approach for maintaining academic integrity among students in the GenAI era. We draw from literature on academic integrity and higher education policies and guidelines to propose eight strategies for maintaining academic integrity among students. These strategies relate to the three stakeholders of students, educators, and institutions, who play a critical role in maintaining academic integrity in HEIs (Khan, 2024; Kier & Ives, 2022). For students, we propose that emphasizing collaborative learning, enhancing critical thinking skills, and improving AI literacy to reduce the reliance on GenAI tools. This approach promotes their responsible use in the learning process while maintaining the academic integrity in assessments. Educators can adopt innovative pedagogical and assessment approaches, such as project-based assessments, authentic assessments, adaptive learning, and participate in development programs to encourage authentic student learning and be updated on the recent developments in GenAI. Finally, institutions must develop technological infrastructure, have clear policies on the use of GenAI tools, and create a culture of academic integrity to discourage students from engaging in academic dishonesty practices during the assessments. Such a comprehensive strategy involving the key stakeholders in HEI can lead to shared commitment to integrity and facilitate development of academic integrity culture, thereby promoting responsible use of GenAI tools in their educational journey.

Our paper is conceptual in nature and is based on the comprehensive relevant literature, including academic papers and institutional documents. This approach was chosen to provide an in-depth understanding of the multifaceted challenges and opportunities that GenAI tools present to academic integrity in HEIs. By incorporating key literature and real-world practices, we provide a holistic framework of academic integrity framework for students in the GenAI era. To the best of our knowledge, this is one of the initial studies to adopt a stakeholder approach to academic integrity, considering the perspectives from students, institutions, and educators.

The rest of the article is structured as follows. We discuss the role of GenAI in HEI, academic integrity, assessment practices, and university policies in the following sections. Next, we forward eight strategies concerning the three stakeholders for maintaining academic integrity. Finally, we conclude with the discussion, challenges, and directions for future research.

2. Literature review

2.1. GenAI in HEI

GenAI represents a new paradigm in the Web 3.0 environment (Dwivedi et al., 2023). It is an umbrella term used to describe machine learning solutions or models. GenAI is characterized by its ability to generate new and original data or content based on prompt commands (Sætra, 2023). Along with the ability to analyze and interpret existing information that previous generations of AI

tools could do, GenAI can generate new data or content across various forms, including text, images, and audio. The key distinction lies in creating new content with higher efficiency and greater creativity that meets or even exceeds human output but at a fraction of the cost (Dwivedi et al., 2024; Epstein et al., 2023). The key component of GenAI is the Natural Language Processing (NLP) model that allows it to understand, interpret, and generate content from the training data (Sætra, 2023). NLP enables GenAI tools to create meaningful, natural, and personalized content, transforming the user experience into a more dynamic and interactive process (Chaisatitkul et al., 2024; Ooi et al., 2023).

The introduction of GenAI tools has triggered considerable debate among stakeholders of HEIs. Advocates of GenAI tools argue that they can transform the educational landscape by personalizing content for specific student needs, increasing accessibility to information and knowledge, and providing more interactive learning opportunities for students (Baidoo-Anu & Ansah, 2023). Furthermore, it shifts students from passive receivers of information to active participants in the learning process (Ratten & Jones, 2023). Additionally, it can offer tailored responses to student queries on diverse topics in real-time, thereby enhancing their learning experience (Peres et al., 2023). It can help educators analyze student data and identify their distinct learning styles. This helps in personalizing the learning content and experience. Despite these benefits, some have raised alarm bells regarding its adverse impact on students' learning outcomes. The made-up data, biases, and hallucinations of GenAI tools impact the accuracy and authenticity of the information that students use in their learning process (Monteith et al., 2024). Additionally, concerns related to privacy, intellectual property, and copyright have forced HEIs to critically investigate how GenAI tools impact students' academic integrity (Adeshola & Adepoju, 2023).

2.2. GenAI and academic integrity

Maintaining academic integrity is the cornerstone of higher education (Sefcik et al., 2019). Student behaviors, such as plagiarism, dishonesty, fraud, and cheating, constitute violations of academic integrity (Jones, 2011). These violations hinder accurate assessment of students' knowledge and abilities. It further questions the fairness of the educational process (Awosoga et al., 2021). Academic integrity is not confined to only student behaviors but also involves the conduct of the academic community that is responsible for implementing standards, promoting awareness, and fostering discussions on academic integrity (Caldwell, 2010). Moreover, it can develop a learning environment that values integrity and honesty (Bin-Nashwan et al., 2023). This indicates that academic integrity is a commitment to maintaining honesty, responsibility, and trust by all those involved in the HEIs, including students, educators, and the administration. However, the introduction of GenAI tools has resulted in significant challenges to academic integrity, requiring a comprehensive approach to uphold it.

A few recent studies have investigated the relationship between GenAI tools and academic integrity of various stakeholders. For example, Bin-Nashwan et al. (2023) found a negative relationship between academic integrity and ChatCPT use among academicians in HEIs. Similarly, Cotton et al. (2024) discussed the potential risk of ChatGPT on academic integrity in academia. Along with these studies, industry report reveal that approximately 46 percent of US higher education students frequently use GenAI tools such as ChatGPT for their homework (Intelligent, 2023). However, approximately 33 percent of students were not aware that these tools can generate false information (Freeman, 2024), and 40 percent of the students reported limited knowledge about the capabilities of GenAI tools (Kelly et al., 2023). While students may view GenAI tools as supportive tools or study companion (Li et al., 2024), they may intentionally or unintentionally compromise academic integrity by relying on them in completing the assessments. This is further complicated by their lack of understanding of GenAI tools and the data or information it generates, leading to academic dishonesty.

2.3. GenAI and assessment approaches

Assessment encompasses testing, grading, and providing feedback, all designed to guide students toward improvement (Heywood, 2000). It can be either formative or summative, depending on its goals; however, both serve distinct purposes in evaluating students' learning outcomes and providing feedback. Formative assessment takes place throughout the students' learning process. It offers immediate feedback, which helps students and educators identify areas of improvement. Furthermore, it enables educators to modify the pedagogical approach to meet specific needs of the student (Taras, 2008). On the other hand, summative assessment is usually conducted at the end of the students' learning process to assess the extent to which the learning outcomes are met. Together, formative and summative assessments provide a comprehensive view of student learning and progress, enabling educators to tailor the instructional approach and offer necessary support to facilitate students in their learning process (Hernández, 2012).

It is therefore not enough to incorporate AI technologies into educational practices at large but rather to consider their specific approaches to assessment. The concept of Generative AI could change over time, as it might place less emphasis on assessment. At the same time, it is critical for educators not to lose sight of important issues and possibilities that AI contributes to the assessment of students' learning. What is important is being able to guarantee that assessments are still valid, fair, and able to determine the extent of the students' knowledge without the help of AI-generated support.

With the rise of GenAI, assessment approaches have become a crucial priority for educators. These assessments should be designed to encourage students to engage in original work and not submit AI-generated content. Specifically, HEIs should develop a learning environment where students are motivated to use their own skills and capabilities in completing their assessments rather than relying on shortcuts such as GenAI tools. Recent research has investigated the assessment approaches of HEIs in the GenAI era. For example, Moorhouse et al. (2023) reviewed Generative AI guidelines on the websites of 23 leading universities. Their findings revealed three main areas of focus: academic integrity, advice on assessment design, and communication with students. The authors recommended that educators should develop effective assessment tasks, assess the extent to which GenAI tools can complete them, and educate and

allow students to properly use GenAI in completing the assessments. Similarly, [Thanh et al. \(2023\)](#) employed Bloom's taxonomy to develop a framework that enables educators to evaluate the effectiveness of GenAI tools in student assessments. They proposed that educators should develop assessment questions based on Bloom's taxonomy and design marking rubrics and guidelines accordingly. They should then input assessment questions and marking rubrics into GenAI tools, check the answers with marking guidelines, and then grade the answers. This would enable educators to better understand GenAI capabilities and evaluate student assessments more effectively. The above studies provide an initial understanding of approaches adopted by educators and HEIs in assessing students' learning in the context of GenAI tools.

2.4. HEI policies and guidelines on the use of GenAI tools

The potential risks of academic dishonesty has let to HEIs develop policies regarding the use of GenAI tools by various stakeholders ([Luo, 2024](#)). While many HEIs have published AI-related guidelines, they have not considered the scope of impact of GenAI tools on student learning. For example, [Plata et al. \(2023\)](#) investigated how top 20 global universities approached the role of GenAI tools on academic integrity and student learning. Their study identified three approaches, namely, education, enforcement, and encouragement, for dealing with the risk of GenAI tools on academic integrity. Australia's The Tertiary Education Quality and Standards Agency (TEQSA) established two foundational principles to guide the creation of assessments in the context of GenAI ([TEQSA, 2023](#)). The first principle states that assessments and learning experiences should prepare students to engage actively and ethically in a society where GenAI is widely used. The second principle emphasizes making reliable judgments about student learning in the GenAI era requires inclusive, contextualized, and diverse assessment methods ([TEQSA, 2023](#)).

Monash University in Australia articulated a cautious stand toward the integration of GenAI tools in student submissions. It advocated for the responsible use of GenAI tools by students in their assessments while emphasizing the need to acknowledge their use ([Monash University, n.d.](#)). Similarly, Harvard University advocated a judicious use of GenAI, highlighting the criticality of safeguarding information security, data privacy, compliance, copyright, and academic integrity ([Harvard University, n.d.](#)). Oxford University echoed a similar sentiment, advocating responsible use of GenAI tools by both faculty and students in their academic pursuits. Additionally, it emphasized to stakeholders that GenAI is not an expedient solution or alternative for the personal commitment needed to foster intellectual aptitude of students ([University of Oxford, n.d.](#)). While these policies and guidelines offer critical insights into how HEIs are dealing with GenAI, they do not offer a comprehensive or inclusive approach for managing academic integrity. [Luo \(2024\)](#) states that most policies of HEIs revolved around 'what works' but they do not focus on the effectiveness of such policies in maintaining academic integrity.

3. A stakeholder approach to maintaining academic integrity in the GenAI era

This study adopted a stakeholder approach to propose a comprehensive academic integrity framework in HEIs. Given the complexities and multifaceted nature of maintaining academic integrity, our approach highlights the need to engage all critical stakeholders in the HEIs. Such an approach can foster an environment for ethical and responsible use of GenAI tools among students and allows accurate assessment of their learning. We developed our framework based on an extensive review of relevant literature and institutional documents detailing policies and guidelines.

We identify eight strategies ([Fig. 1](#)): three focused on students, namely, Collaborative Learning, Enhancing Critical Thinking Skills, and Developing AI Literacy; two focused on educators, namely, Innovative Pedagogical and Assessment Approaches, and Educator Development Programs; and three focused on institutions, namely, Investing in Technological Infrastructure, Developing Comprehensive GenAI Policies, and Fostering a Culture of Academic Integrity. To implement these strategies, we present ten propositions focusing on enhancing collaborative learning, critical thinking, and AI literacy for students; adopting adaptive, authentic, and project-

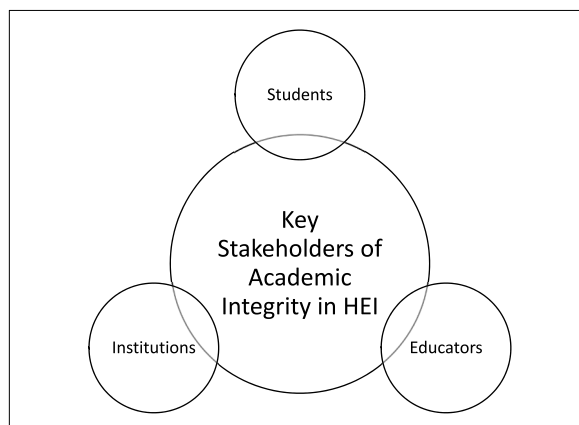


Fig. 1. Stakeholder approach to academic integrity.

based assessments; providing educator training; investing in technological infrastructure; developing comprehensive GenAI policies; and fostering a culture of academic integrity. Along with this, we identify four distinct challenges for implementing the comprehensive academic integrity framework, namely, Strategic Challenges, Operational Challenges, Cultural Challenges and Financial Challenges. We considered the five academic integrity values of honesty, trust, fairness, respect, and responsibility when developing this framework (International Center for Academic Integrity, 2021). These values are integral to fostering academic honesty among stakeholders and encourage students to responsibly use GenAI tools in their learning process. In the following sections, we discuss these strategies and describe the process by which they contribute to maintaining academic integrity among students in the GenAI era.

4. Empowering students for integrity

Students are key stakeholders in HEI for maintaining academic integrity. Specifically, they have a critical role in promoting ethical standards with their academic work and scholarly pursuits. Given this, we propose three strategies namely, engaging in collaborative learning, enhancing critical thinking skills, and developing AI literacy that are essential for upholding academic integrity and honesty among students.

4.1. Collaborative learning

Collaborative learning is an educational approach that encourages students to work together on tasks or assessments toward a common goal (Vuopala et al., 2016). It is grounded in the sociocultural theory, which suggests that social interaction plays a key role in the cognitive development of students (Hmelo-Silver et al., 2008). Collaborative learning emphasizes social aspects where students can support one another in the learning process, thereby achieving a higher level of understanding through collaborative interactions among students. Previous research has demonstrated that collaborative learning creates a sense of belongingness, enhances diversity, and prepares students for the challenges of the real world (Seifert & Bar-Tal, 2023). Furthermore, researchers have shown that collaborative learning increases student performance and learning satisfaction (Cen et al., 2016; Cheng et al., 2021). As collaborative learning creates a dynamic social environment where students work together, they not only share knowledge but also challenge each other's assumptions and understanding, leading to an in-depth understanding of the subject matter (Dewiyanti et al., 2007).

We propose that engaging in collaborative learning helps maintain academic integrity among students. Collaborative learning promotes core values of academic integrity, such as transparency, responsibility, and fairness, by making students aware that their contributions are subject to peer review and collective scrutiny (Kotsonis, 2022). It also nurtures a culture of honesty and shared accountability. Such shared responsibility enable students to recognize that their contributions are pivotal for the overall success of the group. By collaborating together, students can leverage the strength and perspectives of others in the group (Chauhan et al., 2018). By encouraging shared responsibility and trust among group members, collaborative learning approach plays a key role in maintaining academic integrity among students in the GenAI era. For example, Stanford University promotes a collaborative environment to make students actively involved and engaged in their learning process (Standard Graduate School of Business, n.d.). Similarly, The University of Melbourne's Peer Assisted Study Sessions aims to maximize student engagement and develop a sense of community (The University of Melbourne, n.d.). The above discussion suggests that promoting collaborative learning among students, HEI can promote academic honesty among students in the GenAI era.

Proposition 1. *Collaborative learning strategies are crucial to maintaining academic integrity among students in the GenAI era.*

4.2. Enhancing critical thinking skills

Critical thinking is a key learning outcome for students in HEIs. It is critical for the overall development and academic success of students (Stupnisky et al., 2008). Critical thinking skills involve the ability of the students to think rationally and critically and make informed decisions (Walters, 1990). These skills helps students differentiate facts from opinions, evaluate arguments and evidence, establish logical connections, and systematically solve problems (Tiruneh et al., 2014). Thus, enhancing critical thinking skills prepare students to manage complex problems and engage in lifelong learning.

In the GenAI era, enhancing critical thinking skills in students is even more significant for maintaining academic integrity. As GenAI tools can hallucinate and create inaccurate or incomplete output, students should possess critical thinking skills to evaluate the reliability of content or information generated by GenAI tools. This ensures that students responsibly use GenAI tools in their learning process. For example, the University of Bath recently introduced a new module that focuses on building students' critical thinking skills and confidence in using GenAI tools (University of Bath, 2023). Similarly, the University of Technology Sydney offers various GenAI learning resources aimed at enhancing the students' critical thinking skills (UTS, n.d.). The above discussion indicates that critical thinking is the key to unlocking the potential of GenAI for student learning. It can transform GenAI from mere technological tools to powerful learning aids.

Proposition 2. *Enhancing critical thinking skills is crucial for maintaining academic integrity among students in the GenAI era.*

4.3. Training and supporting students in developing AI literacy

Developing AI literacy in students is becoming increasingly crucial in today's context. It is the ability of students to efficiently use AI tools and have a deeper understanding of their applications, biases, and limitations (Kong et al., 2023). By developing AI-literacy, HEI

can empower students to responsibly use GenAI in their learning process. Kong et al. (2021) proposed an AI literacy framework consisting of three dimensions, namely, cognitive, affective, and sociocultural. The cognitive dimension relates to educating students about the basics of AI and developing their competencies. The affective dimension relates to their empowerment, while the socio-cultural aspects concerns their awareness of ethical considerations in using AI in their learning process. Similarly, Southworth et al. (2023) proposed a model of AI literacy that encompasses enabling AI, knowing and understanding AI, using and applying AI, evaluating and creating AI, and AI ethics dimensions. The authors propose that AI literacy programs should be aligned with student learning outcomes, as it can foster more engaged students and transform them into digital citizens. Furthermore, it results in higher levels of AI self-efficacy, awareness, and empowerment (Kong et al., 2023).

In the present study, we propose that enhancing AI literacy in students are pivotal for maintaining academic integrity. AI literacy provides a comprehensive understanding of AI tools and their ethical use. By increasing the technical competence of GenAI tools, it enable students to distinguish between responsible and unethical use of GenAI in their assessments. For example, The University of Sydney offers its students various resources to help them upskill in their use of GenAI tools (Liu & Bridgeman, 2023). RMIT University recently launched a module for students across different disciplines, focusing on the university policies, guidelines, and resources related to GenAI and academic integrity (RMIT, 2024). Similarly, Nanjing University launched an AI general curriculum that covers the knowledge, skills, values, and ethics of AI use, aiming to enhance AI literacy among students (Global Times, 2024). Thus, enhancing AI literacy ensures that students are technically proficient and ethically grounded in their use of GenAI tools in their learning process.

Proposition 3. *AI literacy programs are crucial for maintaining academic integrity among students in the GenAI era.*

5. Equipping educators for academic integrity

Along with students, this study identifies educators as another key stakeholder in maintaining academic integrity among students in the GenAI era. Recognizing the critical role educators play in maintaining a learning-centric environment, we propose two key strategies for maintaining academic integrity among students. The first strategy focuses on adoption of innovative pedagogical and assessment approaches that discourages academic dishonesty. The second approach emphasizes the importance of comprehensive educational training programs to ensure educators possess necessary skills for creating a culture of honesty and ethical behaviors.

5.1. Innovative pedagogical and assessment approaches

In this section, we focus on three innovative pedagogical (adaptive learning) and assessment (authentic and project-based) approaches for promoting academic honesty among students in the GenAI era. While other approaches exist, we selected these three approaches due to their ability to make students adopt academic integrity values in their learning process.

5.1.1. Adaptive learning

Adaptive learning is an emerging technological innovation in the higher education context (Smyrнова-Trybulska et al., 2022). It utilizes technologies to create interactive content that dynamically changes based on students' understanding and knowledge (Kerr, 2016). It uses integrated assessments to tailor learning materials and contribute to unique learning experiences for each student. The personalized approach and the feedback system promotes self-paced learning and increases students' confidence and engagement (Contrino et al., 2024). HEIs have adopted various platforms such as McGraw Hill's ALEKS, Alta's Knewton, and Pearson Interactive Labs to promote adaptive learning among students. This study proposes adaptive learning approaches or platforms enhance students' learning efficiency through personalized learning process that aligns with their individual learning styles. These approach increase student engagement by encouraging them to use their own abilities and competencies. Furthermore, they increase students' self-efficacy and foster a sense of ownership and commitment, thereby ensuring that they are less likely to engage in unethical or dishonesty practices (Martin et al., 2020).

Proposition 4. *Adopting adaptive learning platforms is crucial for maintaining academic integrity among students in the GenAI era.*

5.1.2. Authentic assessments

Authentic assessments allow educators to fairly assess the skills and knowledge of students in their learning process. These assessments, such as case studies, simulations, and experiential learning activities, evaluate student learning and knowledge through tasks and activities that are based on real-world situations and scenarios (Hobbins et al., 2022). By simulating real-world situations, these assessments motivate students to apply their actual understanding and knowledge in a more practical and meaningful way. This approach increases students' critical thinking and problem-solving skills (Swaffield, 2011). They bridge the gap between theory and practice and help better prepare students to face modern business world challenges. Given the unique and contextual nature of these assessments, students have to deeply engage with the assessment material. Such engagement shifts students from passive receivers of information to actively constructing knowledge. As GenAI tools may not offer specific solutions for the real-world problem emphasized in the authentic assessments (Burtsev et al., 2023), students are compelled to either avoid them or augment them with their own insights and knowledge. Thus, authentic assessments may help maintain academic integrity through genuine student efforts as well as critical application of GenAI tools. For example, Maynooth University promotes authentic assessment in its programs and even designed a six-step task generator for creating authentic assessments (Maynooth University, n.d.).

Proposition 5. *Utilizing authentic assessments is crucial for maintaining academic integrity among students in the GenAI era.*

5.1.3. Project-based assessments

Another innovative assessment approach educators can utilize to promote academic integrity among students in the GenAI era is project-based assessments. This approach involves students working on a long-term project requiring them to apply their knowledge and skills to address real-world problems. The students are required to ask questions, collect information and data, analyze and interpret findings, and draw conclusions (Tal et al., 2000). Thus, project-based assessments simulate real-world business problems and integrate both theory and practice (Van den Bergh et al., 2006). However, they often involve an industry partner and usually span a longer duration of time whether its outcomes are implemented outside the classroom.

We propose that by using project-based assessments, educators can play a key role in enhancing academic integrity among students in the GenAI era. This is because project-based assessments require students to partner with external parties and make decisions based on real-world problems (Rohm et al., 2021). As GenAI tools may not fully understand the nuances and complexities of the specific problems or challenges faced by businesses, their utility may be limited (Burtsev et al., 2023). While GenAI tools can create new content based on existing data, it lacks the ability to address novel and specific requirements set by external partners in project-based assessments (Rahman & Watanobe, 2023). This makes students rely on their critical thinking, creativity, and problem-solving skills, thereby promoting academic honesty practices among students in project-based assessments. For example, Maastricht University incorporates problem-based learning across its programs to tackle real-life challenges (Maastricht University, n.d.).

Proposition 6. *Employing project-based assessments is crucial for maintaining academic integrity among students in the GenAI era.*

5.2. Educator Development Programs

Effective development programs are critical for preparing educators to address the challenges GenAI poses to academic integrity. These personal and professional development programs equip educators with relevant skills and competencies to successfully guide students in their learning process (Centra, 1978). Educator development programs provide up-to-date knowledge on teaching and learning innovations, effective assessment strategies, and digital literacy (Camblin Jr & Steger, 2000). These workshops also focus on enhancing facilitation skills, such as guiding discussions, managing group dynamics, fostering collaboration, and fostering a student-centric learning environment (Chiu, 2023). Additionally, they promote a growth mindset in educators which enable them to adopt efficient pedagogical approaches for improving student learning (Lin et al., 2022). Specifically, digital literacy programs prepare educators to use GenAI tools effectively, critically evaluate AI-generated content, and develop strategies for instilling academic integrity among students. Thus, educational development programs develop competencies to reshape the pedagogical approach of educators. By offering continuous development programs, HEIs can empower educators with the necessary skills and knowledge to address the challenges of GenAI tools in maintaining academic integrity among students in their classrooms.

Proposition 7. *Educator development programs are crucial for maintaining academic integrity among students in the GenAI era.*

6. Institutional foundations for integrity

This paper delineates three pivotal institutional strategies for maintaining academic integrity among students: the development of a technological infrastructure, implementing a comprehensive GenAI policy, and cultivating a culture of academic integrity. These strategies are essential for fostering an environment where academic honesty practices are encouraged among various stakeholders including students.

6.1. Technological infrastructure

Technological infrastructure encompass the set of digital technologies, tools, and applications deployed in HEIs to facilitate and support the educational and administrative processes for student, educators, and other stakeholders (Blind & Grupp, 1999; Rodríguez-Abitia et al., 2020). Technological infrastructure includes learning management systems, digital libraries, online collaboration platforms, and research-related software (Schneckenberg, 2009). It also covers the necessary hardware and network infrastructure that allows seamless access to digital resources. These technological infrastructure enhances student learning experience, manage operational processes, and foster a community within the academic environment. Previous research has demonstrated that technological infrastructure improves the quality of educational programs and enhances student engagement (Engel et al., 2023; Schindler et al., 2017).

It is noteworthy, however, that some technological tools are somewhat ineffective in identifying the improper utilization of AI. Some institutions have decided not to implement AI detection tools yet due to issues of efficiency and surveillance. For these reasons, a much more sensible approach to the management of cheating must be planned using supervised technologies accompanied by human oversight and ethics to ensure the fairness and integrity of evaluations.

We propose that HEIs should invest in advanced tools and platforms that facilitate the detection and prevention of dishonest practices among students. Plagiarism and AI-detection software enables students to present genuine work and helps to maintain academic integrity. Furthermore, it encourages the development of critical thinking and problem-solving skills. It also prepares students for a future where they can responsibly and ethically use GenAI tools in their ongoing studies. Thus, technological infrastructure forms the foundation of an environment in which students' academic integrity is not compromised.

Proposition 8. *Investing in advanced technological infrastructure is crucial for maintaining academic integrity among students in the GenAI*

era.

6.2. Comprehensive GenAI policy development and review

Implementing a comprehensive AI policy for responsible use of GenAI tools is critical for maintaining academic integrity among students in HEI. This comprehensive policy should include regulations for the use of GenAI tools, clearly communicating the roles and responsibilities of various stakeholders for its responsible use, and outline procedures for detecting academic dishonesty and penalties and sanctions for their use, among others. For example, Aalto University's recent academic integrity policy introduced detailed guidelines for the responsible use of AI tools for students and other stakeholders (Aalto University, 2023). Similarly, the new integrity policy introduced by The University of Sydney offers detailed guidelines for AI use by students in assessable work and outlines the consequences of academic dishonesty (The University of Sydney, 2023). While these examples reveal proactive steps taken by HEI, Luo (2024) criticized current AI policies in various universities for its lack of comprehensiveness and recommends implementing specific guidelines for use of GenAI tools without limiting its use for enhancing student learning. Chan (2023) proposed a comprehensive GenAI policy that includes guidelines for the pedagogical and ethical use of AI tools for maintaining academic integrity. Additionally, the United Nations Educational, Scientific, and Cultural Organization (UNESCO) has issued guidelines advocating the regulation of GenAI in education through human-centric approaches that prioritize its ethical, safe, equitable, and meaningful use (UNESCO, 2023). As GenAI continues to evolve, HEI should develop a comprehensive policy and regularly review it to ensure it remains relevant.

Proposition 9. *A dynamic approach to comprehensive AI policy development and review is crucial for maintaining academic integrity among students in the GenAI era.*

6.3. Culture of academic integrity

We propose that fostering a culture of academic integrity in HEIs is essential to imparting fundamental values such as honesty, integrity, trust, fairness, respect, responsibility, and courage. It also empowers students to achieve success and bring about positive change in communities (Çelik & Razi, 2023). McCabe et al. (2012, p. 3) viewed academic integrity as “a harbinger of things to come, a reflection of the general mores that society is passing on to the next generation.” Thus, fostering a culture of academic integrity is critical as academic dishonesty is a common, contagious, and corrosive epidemic (Çelik & Razi, 2023). Various scholars have proposed models or frameworks for creating a culture of academic integrity. For example, Wangaard and Stephens (2011) proposed a four-dimension model comprising core values, committee and commitments, culture and community, and curriculum and instruction for promoting academic integrity at both school and classroom levels. Similarly, Hossain (2022) proposed a 4P academic integrity model encompassing people, policy, preparation, and practice.

It is essential to note that this culture of academic integrity is not fixed but needs to be dynamic, considering the character of technology. Just as the arrival of the Internet more than twenty years ago sparked a reconsideration of the concept of academic integrity, the application of Generative AI tools should trigger a critical look at the definition of integrity today. Organizations need to not only maintain their culture of integrity but reshape it to respond to new the threats and opportunities presented by AI in all subjects.

Recent studies investigating the implementation of an academic integrity culture indicate that it is a slow and challenging process that requires the active participation of all key stakeholders (Stephens & Wangaard, 2016). They also identified various barriers to developing such a culture, including a lack of response to academic dishonesty incidents, prioritizing academic success over integrity, resistance from educators to change, and traditional assessment approaches such as exams (Çelik & Razi, 2023). Despite these challenges, developing a culture of academic integrity is critical, as GenAI tools can undermine the core values of academic honesty (Duah & McGivern, 2024). Thus, we propose that cultivating a culture of academic integrity ensures that students use GenAI tools responsibly and ethically, ensuring trust, fairness, and transparency in their assessments.

Proposition 10. *Developing a culture of academic integrity is crucial for maintaining academic integrity among students in the GenAI era.*

Each strategy proposed for key stakeholders is an essential part of the comprehensive approach to maintaining academic integrity among students in the GenAI era. While each individual strategy is a valuable tool, their combined implementation, impacting all key stakeholders, is crucial for overall success in promoting students' responsible use of GenAI in their learning process.

7. Discussion

GenAI presents significant opportunities and challenges for students and other stakeholders in HEIs. While it can enhance student learning and engagement through personalized and interactive content, it also raises concerns about academic dishonesty. Given that GenAI tools are evolving, it is crucial for HEIs to develop comprehensive strategies that address the challenges they pose to academic integrity while leveraging them for enhancing their learning process. This study presents a comprehensive academic integrity framework that encompasses eight strategies related to three key stakeholders of students, educators, and institutions. This framework emphasizes collaborative learning, critical thinking, and AI literacy for students, innovative pedagogical and assessment approaches and development programs for educators, and developing comprehensive AI policy and review, culture of academic integrity, and technological infrastructure for institutions (Fig. 2). The proposed framework offers a foundational approach for HEIs to proactively engage with multiple stakeholders to maintain academic integrity among students and ensures that education remains a process that genuinely reflects students' learning involving the ethical use of GenAI tools. However, it is important to note that implementing such a

comprehensive framework is not without its challenges.

7.1. Challenges for academic integrity framework

Implementing the proposed comprehensive academic integrity framework presents significant challenges for HEIs. Recognizing these challenges will help them better prepare for its effective implementation. We identified four key challenges, i.e., financial, operational, strategic, and cultural challenges, each presenting a distinct barrier that requires careful consideration and planning (Fig. 3).

- (1) Strategic challenges: Achieving consensus among all stakeholders on the responsible and ethical use of GenAI tools can be challenging (Duah & McGivern, 2024). Additionally, aligning these strategies with HEIs goals and objectives present significant hurdles. HEIs may also face resistance from various stakeholders demanding their involvement in the policy-making process.
- (2) Operational challenges: Keeping pace with the rapid development of AI tools, including GenAI, requires continuous development and revision of the HEIs policies, pedagogical approaches, and assessment methods. Moreover, increasing AI literacy and equitable access to GenAI resources involve significant operational and support considerations (Chiu, 2023).
- (3) Cultural challenges: Creating a culture of academic integrity not only relates to developing policy and training but also changing the mindset of student to value honesty, fairness, and ethics in their learning process.
- (4) Financial challenges: Significant financial investments are required to invest in the appropriate technological infrastructure to support responsible use of GenAI tools by students in their learning process (Perkins et al., 2023). Furthermore, designing and offering development and training programs for students and educators involves substantial costs.

7.2. Future research directions

As the GenAI landscape continues to evolve, it is important for HEIs to identify the future directions for understanding and addressing both its opportunities and challenges. We identified four key future research directions.

First, HEI could assess the current levels of AI and GenAI literacy among students and educators. These insights would enable HEI to tailor policies and guidelines to meet specific needs that foster the responsible use of GenAI among students. HEIs can adopt AI literacy scales developed by Laupichler et al. (2023) or Carolus et al. (2023) to measure GenAI literacy levels. This ensures an accurate assessment of current levels of GenAI among various stakeholders when developing the academic integrity policy.

Second, HEI could assess the impact of GenAI tools on student learning outcomes. This includes examining whether GenAI enhance problem-solving, critical thinking, creativity, and decision-making skills. While previous research has shown that GenAI tools increase productivity by 14% and reduce task completion time by 40% among workers (Brynjolfsson et al., 2023; Noy & Zhang, 2023), limited research has examined their impact on student competences and skills.

Third, HEI should investigate the role of technological infrastructure in maintaining academic integrity among students. Specifically, future research could focus on identifying relevant and effective tools and platforms for preventing academic dishonesty among students. Similarly, future research could carry out longitudinal studies examining the long-term impact of GenAI on academic integrity.

Finally, future research should investigate how different pedagogical and assessment approaches impact students' academic

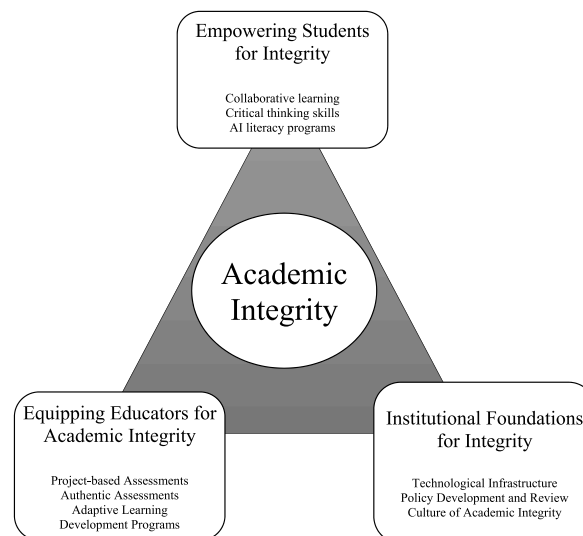


Fig. 2. Academic Integrity Framework in the GenAI era.

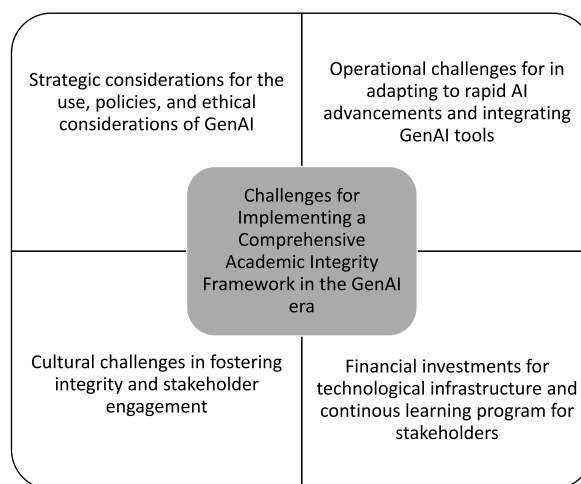


Fig. 3. Challenges for implementing the GenAI Academic Integrity Framework.

integrity. While previous research has demonstrated the effectiveness of authentic assessments and project-based assessments in reinforcing student learning, future research should examine how they promote ethical behaviors and maintains fairness among students.

8. Conclusion

The study presents a comprehensive framework for academic integrity, targeting the key stakeholders in HEIs: students, educators, and institutions. Specifically, it outlines eight key strategies, from fostering collaborative learning among students to developing comprehensive GenAI policy at the institutional level. By adopting this framework, HEIs can effectively address the challenges GenAI poses to academic integrity among students. However, further research is needed to develop a deeper understanding of how this framework can be implemented effectively across diverse educational contexts and cultures.

It is also acknowledged that the impact of GenAI in higher education is changing dynamically. This has made it further difficult to determine the long-term policies. Thus, our paper proposes for the development of dynamic policy to regulate these issues owing to their uncertainty. We understand that such opportunities as, for example, the use of GenAI tools leads to hallucination in the future, may be minimized, and there may be other concerns. Therefore, the proposed framework provides an opportunity for the constant tracking and updating of the existing interventions as a means of preserving their efficacy. It will also help HEIs in ensuring that it is ready for the new future changes and trends in GenAI as the institution upholds its academic integrity.

In addition, it is important to note that there is a clear need to relate the assessment strategies with reference to the GenAI. Authentic and project-based assessments both have been considered as best practices and their significance has increased due to the emergence of GenAI tools. These methods reduce the possibilities of misuse of AI by enforcing the need for more personal interactions with the AI. Thus, GenAI tools support the effectiveness of these strategies and encourages educators to think beyond traditional exams and written assignments. In response to the possible change of moving back to invigilated exams, our framework suggest the designing of assessments that are more suited to GenAI but at the same time uphold academic integrity through critical thinking.

Declaration of AI-assisted technology in the writing process

During the preparation of this work, the authors used Grammarly, a language editing software, for language suggestions and copy-editing assistance in some parts of this article. After using this tool, the authors reviewed the content as needed and take full responsibility for the content of the publication.

Funding

None.

CRedit authorship contribution statement

Tareq Rasul: Writing – review & editing, Writing – original draft, Visualization, Supervision, Project administration, Conceptualization. **Sumesh Nair:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Project administration, Conceptualization. **Diane Kalendra:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Project administration, Conceptualization. **M.S. Balaji:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Project

administration, Conceptualization. **Fernando de Oliveira Santini:** Writing – review & editing, Writing – original draft, Visualization, Conceptualization. **Wagner Junior Ladeira:** Writing – review & editing, Writing – original draft, Visualization, Conceptualization. **Raouf Ahmad Rather:** Writing – review & editing, Writing – original draft, Conceptualization. **Naveed Yasin:** Writing – review & editing, Writing – original draft, Visualization, Conceptualization. **Raul V. Rodriguez:** Writing – review & editing, Writing – original draft, Visualization, Conceptualization. **Panagiotis Kokkalis:** Writing – review & editing, Writing – original draft, Visualization, Conceptualization. **Md Wahid Murad:** Writing – review & editing, Writing – original draft, Conceptualization. **Md Uzir Hossain:** Writing – review & editing, Writing – original draft, Visualization, Conceptualization.

Data availability

No data was used for the research described in the article.

Acknowledgments

None.

References

- Aalto University. (2023). Guidance for the use of artificial intelligence in teaching and learning at Aalto University. <https://www.aalto.fi/en/services/guidance-for-the-use-of-artificial-intelligence-in-teaching-and-learning-at-aalto-university>.
- Abdaljaleel, M., Barakat, M., Alsanafi, M., Salim, N. A., Abazid, H., Malaeb, D., Mohammed, A. H., Hassan, B. A. R., Wayyes, A. M., Farhan, S. S., El Khatib, S., Rahal, M., Sahban, A., Abdelaziz, D. H., Mansour, N. O., AlZayer, R., Khalil, R., Fekih-Romdhane, F., Hallit, R., ... Sallam, M. (2024). A multinational study on the factors influencing university students' attitudes and usage of ChatGPT. *Scientific Reports*, 14(1), 1983. <https://doi.org/10.1038/s41598-024-52549-8>
- Adeshola, I., & Adepoju, A. P. (2023). The opportunities and challenges of ChatGPT in education. *Interactive Learning Environments*, 1–14. <https://doi.org/10.1080/10494820.2023.2253858>
- Adigizel, T., Kaya, M. H., & Cansu, F. K. (2023). Revolutionizing education with AI: Exploring the transformative potential of ChatGPT. *Contemporary Educational Technology*, 15(3). <https://doi.org/10.30935/cedtech/13152>
- Awosoga, O., Nord, C. M., Varsanyi, S., Barley, R., & Meadows, J. (2021). Student and faculty perceptions of, and experiences with, academic dishonesty at a medium-sized Canadian university. *International Journal for Educational Integrity*, 17, 1–26. <https://doi.org/10.1007/s40979-021-00090-w>
- Baidoo-Anu, D., & Ansah, L. O. (2023). Education in the era of generative artificial intelligence (AI): Understanding the potential benefits of ChatGPT in promoting teaching and learning. *Journal of AIDS*, 7(1), 52–62. <https://doi.org/10.61969/jai.1337500>
- Bin-Nashwan, S. A., Sadallah, M., & Bouteraa, M. (2023). Use of ChatGPT in academia: Academic integrity hangs in the balance. *Technology in Society*, 75, Article 102370. <https://doi.org/10.1016/j.techsoc.2023.102370>
- Blind, K., & Grupp, H. (1999). Interdependencies between the science and technology infrastructure and innovation activities in German regions: Empirical findings and policy consequences. *Research Policy*, 28(5), 451–468. [https://doi.org/10.1016/S0048-7333\(99\)00007-4](https://doi.org/10.1016/S0048-7333(99)00007-4)
- Brynjolfsson, E., Li, D., & Raymond, L. R. (2023). *Generative AI at work (report No. 31161)*. National Bureau of Economic Research. <https://doi.org/10.3386/w31161>
- Burtsev, M., Reeves, M., & Job, A. (2023). The working limitations of large language models. *MIT Sloan Management Review*, 65(2), 8–10.
- Caldwell, C. (2010). A ten-step model for academic integrity: A positive approach for business schools. *Journal of Business Ethics*, 92, 1–13. <https://doi.org/10.1007/s10551-009-0144-7>
- Camblin Jr, L. D., & Steger, J. A. (2000). Rethinking faculty development. *Higher Education*, 39(1), 1–18. <https://doi.org/10.1023/A:1003827925543>
- Carolus, A., Koch, M. J., Straka, S., Latoschik, M. E., & Wienrich, C. (2023). MAILS-Meta AI literacy scale: Development and testing of an AI literacy questionnaire based on well-founded competency models and psychological change-and meta-competencies. *Computers in Human Behavior: Artificial Humans*, 1(2), Article 100014. <https://doi.org/10.1016/j.chbah.2023.100014>
- Çelik, Ö., & Razi, S. (2023). Facilitators and barriers to creating a culture of academic integrity at secondary schools: An exploratory case study. *International Journal for Educational Integrity*, 19(1), 4. <https://doi.org/10.1007/s40979-023-00125-4>
- Cen, L., Ruta, D., Powell, L., Hirsch, B., & Ng, J. (2016). Quantitative approach to collaborative learning: Performance prediction, individual assessment, and group composition. *International Journal of Computer-Supported Collaborative Learning*, 11, 187–225. <https://doi.org/10.1007/s11412-016-9234-6>
- Centra, J. A. (1978). Types of faculty development programs. *The Journal of Higher Education*, 49(2), 151–162. <https://doi.org/10.1080/00221546.1978.11776610>
- Chaisatitkul, A., Luangngamkhun, K., Noulpum, K., & Kerdvibulvech, C. (2024). The power of AI in marketing: Enhancing efficiency and improving customer perception through AI-generated storyboards. *International Journal of Information Technology*, 16(1), 137–144. <https://doi.org/10.1007/s41870-023-01661-5>
- Chan, C. K. Y. (2023). A comprehensive AI policy education framework for university teaching and learning. *International Journal of Educational Technology in Higher Education*, 20(1), 38. <https://doi.org/10.1186/s41239-023-00408-3>
- Chauhan, P. K., Wood, E., Plummer, T., & Forsyth, G. (2018). Peer-based interventions on academic integrity: Assessing immediate and long term learning. *Journal of Academic Ethics*, 16, 133–149. <https://doi.org/10.1007/s10805-017-9296-8>
- Cheng, F. F., Wu, C. S., & Su, P. C. (2021). The impact of collaborative learning and personality on satisfaction in innovative teaching context. *Frontiers in Psychology*, 12, Article 713497. <https://doi.org/10.3389/fpsyg.2021.713497>
- Chiu, T. K. (2023). The impact of generative AI (GenAI) on practices, policies and research direction in education: A case of ChatGPT and midjourney. *Interactive Learning Environments*, 1–17. <https://doi.org/10.1080/10494820.2023.2253861>
- Contrino, M. F., Reyes-Millán, M., Vázquez-Villegas, P., & Membrillo-Hernández, J. (2024). Using an adaptive learning tool to improve student performance and satisfaction in online and face-to-face education for a more personalized approach. *Smart Learning Environments*, 11(1), 6. <https://doi.org/10.1186/s40561-024-00292-y>
- Cotton, D. R., Cotton, P. A., & Shipway, J. R. (2024). Chatting and cheating: Ensuring academic integrity in the era of ChatGPT. *Innovations in Education & Teaching International*, 61(2), 228–239. <https://doi.org/10.1080/14703297.2023.2190148>
- Crompton, H., & Burke, D. (2023). Artificial intelligence in higher education: The state of the field. *International Journal of Educational Technology in Higher Education*, 20(1), 22. <https://doi.org/10.1186/s41239-023-00392-8>
- Dewiyanti, S., Brand-Gruwel, S., Jochems, W., & Broers, N. J. (2007). Students' experiences with collaborative learning in asynchronous computer-supported collaborative learning environments. *Computers in Human Behavior*, 23(1), 496–514. <https://doi.org/10.1016/j.chb.2004.10.021>
- Duah, J. E., & McGivern, P. (2024). How generative artificial intelligence has blurred notions of authorial identity and academic norms in higher education, necessitating clear university usage policies. *The International Journal of Information and Learning Technology*, 41(2), 180–193. <https://doi.org/10.1108/IJILT-11-2023-0213>
- Dwivedi, Y. K., Kshetri, N., Hughes, L., Slade, E. L., Jeyaraj, A., Kar, A. K., Baabdullah, A. M., Koohang, A., Raghavan, V., Ahuja, M., Albanna, H., Albashrawi, M. A., Al-Busaidi, A. S., Balakrishnan, J., Barlette, Y., Basu, S., Bose, I., Brooks, L., Buhalis, D., ... Wright, R. (2023). “So what if ChatGPT wrote it?” Multidisciplinary perspectives on opportunities, challenges and implications of generative conversational AI for research, practice and policy. *International Journal of Information Management*, 71, Article 102642. <https://doi.org/10.1016/j.ijinfomgt.2023.102642>

- Dwivedi, Y. K., Pandey, N., Currie, W., & Micu, A. (2024). Leveraging ChatGPT and other generative artificial intelligence (AI)-based applications in the hospitality and tourism industry: Practices, challenges and research agenda. *International Journal of Contemporary Hospitality Management*, 36(1), 1–12. <https://doi.org/10.1108/IJCHM-05-2023-0686>
- Engel, O., Zimmer, L. M., Lörz, M., & Mayweg-Paus, E. (2023). Digital studying in times of COVID-19: Teacher-and student-related aspects of learning success in German higher education. *International Journal of Educational Technology in Higher Education*, 20(1), 12. <https://doi.org/10.1186/s41239-023-00382-w>
- Epstein, Z., Hertzmann, A., Akten, M., Farid, H., Fjeld, J., Frank, M. R., Groh, M., Herman, L., Leach, N., Mahari, R., Pentland, A. S., Russakovsky, O., Schroeder, H., & Smith, A. (2023). Art and the science of generative AI. *Science*, 380(6650), 1110–1111. <https://doi.org/10.1126/science.adh4451>
- Farrokhnia, M., Banihashem, S. K., Noroozi, O., & Wals, A. (2023). A SWOT analysis of ChatGPT: Implications for educational practice and research. *Innovations in Education & Teaching International*, 61(3), 460–474. <https://doi.org/10.1080/14703297.2023.2195846>
- Feuerriegel, S., Hartmann, J., Janiesch, C., & Zschech, P. (2024). Generative AI. *Business & Information Systems Engineering*, 66(1), 111–126. <https://doi.org/10.1007/s12599-023-00834-7>
- Freeman, J. (2024). New HEPI policy note finds more than half of students have used generative AI for help on assessments – but only 5% likely to be using AI to cheat. *HEPI*. <https://www.hepi.ac.uk/2024/02/01/new-hepi-policy-note-finds-more-than-half-of-students-have-used-generative-ai-for-help-on-assessments-but-only-5-likely-to-be-using-ai-to-cheat/>
- Fui-Hoon Nah, F., Zheng, R., Cai, J., Siau, K., & Chen, L. (2023). Generative AI and ChatGPT: Applications, challenges, and AI-human collaboration. *Journal of Information Technology Case and Application Research*, 25(3), 277–304. <https://doi.org/10.1080/15228053.2023.2233814>
- Global Times. (2024). China's Nanjing University launches groundbreaking AI core curriculum for new students. *Global Trade*. <https://www.globaltimes.cn/page/202402/1307809.shtml>
- Harvard University. (n.d.). Initial guidelines for the use of generative AI tools at Harvard. <https://huit.harvard.edu/ai/guidelines>
- Herbold, S., Hautli-Janisz, A., Heuer, U., Kikteva, Z., & Trautsch, A. (2023). A large-scale comparison of human-written versus ChatGPT-generated essays. *Scientific Reports*, 13(1), Article 18617. <https://doi.org/10.1038/s41598-023-45644-9>
- Hernández, R. (2012). Does continuous assessment in higher education support student learning? *Higher Education*, 64, 489–502. <https://doi.org/10.1007/s10734-012-9506-7>
- Heywood, J. (2000). *Assessment in higher education: Student learning, teaching, programmes and institutions* (Vol. 56). Jessica Kingsley Publishers.
- Hmelo-Silver, C. E., Chernobitsky, E., & Jordan, R. (2008). Understanding collaborative learning processes in new learning environments. *Instructional Science*, 36, 409–430. <https://doi.org/10.1007/s11251-008-9063-8>
- Hobbins, J., Kerrigan, B., Farjam, N., Fisher, A., Houston, E., & Ritchie, K. (2022). Does a classroom-based curriculum offer authentic assessments? A strategy to uncover their prevalence and incorporate opportunities for authenticity. *Assessment & Evaluation in Higher Education*, 47(8), 1259–1273. <https://doi.org/10.1080/02602938.2021.2009439>
- Hossain, Z. (2022). Ethical literacy education in K-12 using the 4P academic integrity literacy model [Paper presentation]. In *International association for school librarianship conference (IASL2022)*. South Carolina.
- Intelligent. (2023). Nearly 1 in 3 college students have used ChatGPT on written assignments. <https://www.intelligent.com/nearly-1-in-3-college-students-have-used-chatgpt-on-written-assignments/>
- International Center for Academic Integrity (ICAI). (2021). *The fundamental values of academic integrity* (3rd ed.). ICAI.
- Jones, D. L. (2011). Academic dishonesty: Are more students cheating? *Business Communication Quarterly*, 74(2), 141–150. <https://doi.org/10.1177/1080569911404059>
- Kelly, A., Sullivan, M., & Strampel, K. (2023). Generative artificial intelligence: University student awareness, experience, and confidence in use across disciplines. *Journal of University Teaching and Learning Practice*, 20(6), 12. <https://doi.org/10.53761/1.20.6.12>
- Kerr, P. (2016). Adaptive learning. *ELT Journal*, 70(1), 88–93. <https://doi.org/10.1093/elt/ccv055>
- Khan, Z. R. (2024). Academic integrity training module for academic stakeholders: IEPAR framework. *Journal of Academic Ethics*, 22, 9–31. <https://doi.org/10.1007/s10805-024-09517-8>
- Kier, C. A., & Ives, C. (2022). Recommendations for a balanced approach to supporting academic integrity: Perspectives from a survey of students, faculty, and tutors. *International Journal for Educational Integrity*, 18(1), 22. <https://doi.org/10.1007/s40979-022-00116-x>
- Kong, S. C., Cheung, W. M. Y., & Zhang, G. (2021). Evaluation of an artificial intelligence literacy course for university students with diverse study backgrounds. *Computers in Education: Artificial Intelligence*, 2, Article 100026. <https://doi.org/10.1016/j.caeai.2021.100026>
- Kong, S. C., Cheung, W. M. Y., & Zhang, G. (2023). Evaluating an artificial intelligence literacy programme for developing university students' conceptual understanding, literacy, empowerment and ethical awareness. *Educational Technology & Society*, 26(1), 16–30.
- Kotsonis, A. (2022). Educating for collaboration: A virtue education approach. *Ethics and Education*, 17(3), 311–323. <https://doi.org/10.1080/17449642.2022.2111485>
- Laupichler, M. C., Aster, A., Haverkamp, N., & Raupach, T. (2023). Development of the “scale for the assessment of non-experts’ AI literacy”—An exploratory factor analysis. *Computers in Human Behavior Reports*, 12, Article 100338. <https://doi.org/10.1016/j.chbr.2023.100338>
- Lee, V. R., Pope, D., Miles, S., & Zarate, R. (2024). Cheating in the age of generative AI: A high school survey study of cheating behaviors before and after the release of ChatGPT. *Computers in Education: Artificial Intelligence*, Article 100253. <https://doi.org/10.1016/j.caeai.2024.100253>
- Li, T., Ji, Y., & Zhan, Z. (2024). Expert or machine? Comparing the effect of pairing student teacher with in-service teacher and ChatGPT on their critical thinking, learning performance, and cognitive load in an integrated-STEM course. *Asia Pacific Journal of Education*, 44(1), 45–60. <https://doi.org/10.1080/02188791.2024.2305163>
- Lim, W. M., Gunasekara, A., Pallant, J. L., Pallant, J. I., & Pechenkina, E. (2023). Generative AI and the future of education: Ragnarök or reformation? A paradoxical perspective from management educators. *International Journal of Management in Education*, 21(2), Article 100790. <https://doi.org/10.1016/j.ijme.2023.100790>
- Lin, W., Yin, H., & Liu, Z. (2022). The roles of transformational leadership and growth mindset in teacher professional development: The mediation of teacher self-efficacy. *Sustainability*, 14(11), 6489. <https://doi.org/10.3390/su14116489>
- Liu, H., Azam, M., Bin Naeem, S., & Faiola, A. (2023). An overview of the capabilities of ChatGPT for medical writing and its implications for academic integrity. *Health Information and Libraries Journal*, 40(4), 440–446. <https://doi.org/10.1111/hir.12509>
- Liu, D., & Bridgeman, A. (2023). *Supporting students to use AI resources responsibly and productively*. The University of Sydney. <https://educational-innovation.sydney.edu.au/teaching@sydney/supporting-students-to-use-ai-responsibly-and-productively/>
- Luo, J. (2024). A critical review of GenAI policies in higher education assessment: A call to reconsider the “originality” of students’ work. *Assessment & Evaluation in Higher Education*, 1–14. <https://doi.org/10.1080/02602938.2024.2309963>
- Maastricht University. (n.d.). Project-based learning. <https://www.maastrichtuniversity.nl/education/why-um/problem-based-learning>
- Martin, F., Chen, Y., Moore, R. L., & Westine, C. D. (2020). Systematic review of adaptive learning research designs, context, strategies, and technologies from 2009 to 2018. *Educational Technology Research & Development*, 68, 1903–1929. <https://doi.org/10.1007/s11423-020-09793-2>
- Maynooth University. (n.d.). Authentic assessment. <https://www.maynoothuniversity.ie/centre-teaching-and-learning/hub/authentic-assessment>
- McCabe, D. L., Butterfield, K. D., & Trevino, L. K. (2012). *Cheating in college: Why students do it and what educators can do about it*. JHU Press.
- Monash University. (n.d.). Learning and teaching: Teach HQ: AI and assessment. <https://www.monash.edu/learning-teaching/teachhq/Teaching-practices/artificial-intelligence/generative-ai-and-assessment>
- Monteith, S., Glenn, T., Geddes, J. R., Whybrow, P. C., Achtyes, E., & Bauer, M. (2024). Artificial intelligence and increasing misinformation. *The British Journal of Psychiatry*, 224(2), 33–35. <https://doi.org/10.1192/bjp.2023.136>
- Moorhouse, B. L., Yeo, M. A., & Wan, Y. (2023). Generative AI tools and assessment: Guidelines of the world's top-ranking universities. *Computers and Education Open*, 5, Article 100151. <https://doi.org/10.1016/j.caeo.2023.100151>
- Newton, P., & Xiomeriti, M. (2023). ChatGPT performance on multiple choice question examinations in higher education. A pragmatic scoping review. *Assessment & Evaluation in Higher Education*, 1–18. <https://doi.org/10.1080/02602938.2023.2299059>

- Noy, S., & Zhang, W. (2023). Experimental evidence on the productivity effects of generative artificial intelligence. *Science*, 381(6654), 187–192. <https://doi.org/10.1126/science.adh2586>
- Ooi, K. B., Tan, G. W. H., Al-Emran, M., Al-Sharafi, M. A., Capatina, A., Chakraborty, A., Dwivedi, Y. K., Huang, T. L., Kar, A. K., Lee, V. H., Loh, X. M., Micu, A., Mikalef, P., Mogaji, E., Pandey, N., Raman, R., Rana, N. P., Sarker, P., Sharma, A., & Wong, L. W. (2023). The potential of generative artificial intelligence across disciplines: Perspectives and future directions. *Journal of Computer Information Systems*, 1–32. <https://doi.org/10.1080/08874417.2023.2261010>
- Peres, R., Schreier, M., Schweidel, D., & Sorescu, A. (2023). On ChatGPT and beyond: How generative artificial intelligence may affect research, teaching, and practice. *International Journal of Research in Marketing*, 40(2), 269–275. <https://doi.org/10.1016/j.ijresmar.2023.03.001>
- Perkins, M., Roe, J., Postma, D., McGaughan, J., & Hickerson, D. (2023). Detection of GPT-4 generated text in higher education: Combining academic judgement and software to identify generative AI tool misuse. *Journal of Academic Ethics*, 22(1), 89–113. <https://doi.org/10.1007/s10805-023-09492-6>
- Plata, S., De Guzman, M. A., & Quesada, A. (2023). Emerging research and policy themes on academic integrity in the age of Chat GPT and generative AI. *Asian Journal of University Education*, 19(4), 743–758. <https://doi.org/10.24191/ajue.v19i4.24697>
- Rahman, M. M., & Watanabe, Y. (2023). ChatGPT for education and research: Opportunities, threats, and strategies. *Applied Sciences*, 13(9), 5783. <https://doi.org/10.3390/app1309578>
- Rasul, T., Nair, S., Kalendra, D., Robin, M., de Oliveira Santini, F., Ladeira, W. J., Sun, M., Day, I., Rather, R. A., & Heathcote, L. (2023). The role of ChatGPT in higher education: Benefits, challenges, and future research directions. *Journal of Applied Learning and Teaching*, 6(1).
- Ratten, V., & Jones, P. (2023). Generative artificial intelligence (ChatGPT): Implications for management educators. *International Journal of Management in Education*, 21(3), Article 100857. <https://doi.org/10.1016/j.ijme.2023.100857>
- RMIT University. (2024, February 22). Generative AI for students at RMIT [Module]. RMIT University. https://rmit-catalog.rmit.edu.au/browse/catalog/rmit-library/courses/gen-ai-rmit?gl=1*zaqhz* gcl_au*MTAyNzY4ODcxNj4xNzEwNjE1OTA4
- Rodríguez-Abitia, G., Martínez-Pérez, S., Ramírez-Montoya, M. S., & Lopez-Caudana, E. (2020). Digital gap in universities and challenges for quality education: A diagnostic study in Mexico and Spain. *Sustainability*, 12(21), 9069. <https://doi.org/10.3390/su1221906>
- Roe, J., & Perkins, M. (2023). What they're not telling you about ChatGPT: Exploring the discourse of AI in UK news media headlines. *Humanities and Social Sciences Communications*, 10(1), 1–9. <https://doi.org/10.1057/s41599-023-02282-w>
- Rohm, A. J., Steff, M., & Ward, N. (2021). Future proof and real-world ready: The role of live project-based learning in students' skill development. *Journal of Marketing Education*, 43(2), 204–215. <https://doi.org/10.1177/02734753211001409>
- Sætra, H. S. (2023). Generative AI: Here to stay, but for good? *Technology in Society*, 75, Article 102372. <https://doi.org/10.1016/j.techsoc.2023.102372>
- Schindler, L. A., Burkholder, G. J., Morad, O. A., & Marsh, C. (2017). Computer-based technology and student engagement: A critical review of the literature. *International Journal of Educational Technology in Higher Education*, 14, 1–28. <https://doi.org/10.1186/s41239-017-0063-0>
- Schneckenberg, D. (2009). Understanding the real barriers to technology-enhanced innovation in higher education. *Educational Research*, 51(4), 411–424. <https://doi.org/10.1080/00131880903354741>
- Sefcik, L., Striepe, M., & Yorke, J. (2019). Mapping the landscape of academic integrity education programs: What approaches are effective? *Assessment & Evaluation in Higher Education*, 45(1), 30–43. <https://doi.org/10.1080/02602938.2019.1604942>
- Seifert, T., & Bar-Tal, S. (2023). Student-teachers' sense of belonging in collaborative online learning. *Education and Information Technologies*, 28(7), 7797–7826. <https://doi.org/10.1007/s10639-022-11498-3>
- Smyrnova-Trybulska, E., Morze, N., & Varchenko-Trotsenko, L. (2022). Adaptive learning in university students' opinions: Cross-border research. *Education and Information Technologies*, 27(5), 6787–6818. <https://doi.org/10.1007/s10639-021-10830-7>
- Southworth, J., Migliaccio, K., Glover, J., Reed, D., McCarty, C., Brendemuhl, J., & Thomas, A. (2023). Developing a model for AI across the curriculum: Transforming the higher education landscape via innovation in AI literacy. *Computers in Education: Artificial Intelligence*, 4, Article 100127. <https://doi.org/10.1016/j.caeai.2023.100127>
- Stanford Graduate School of Business. (n.d.). The experience: Collaborative environment. <https://www.gsb.stanford.edu/experience/life/collaborative-environment>
- Stephens, J. M., & Wangaard, D. B. (2016). The achieving with integrity seminar: An integrative approach to promoting moral development in secondary school classrooms. *International Journal for Educational Integrity*, 12, 1–16. <https://doi.org/10.1007/s40979-016-0010-1>
- Stupnisky, R. H., Renaud, R. D., Daniels, L. M., Haynes, T. L., & Perry, R. P. (2008). The interrelation of first-year college students' critical thinking disposition, perceived academic control, and academic achievement. *Research in Higher Education*, 49, 513–530. <https://doi.org/10.1007/s11162-008-9093-8>
- Swaffield, S. (2011). Getting to the heart of authentic assessment for learning. *Assessment in Education: Principles, Policy & Practice*, 18(4), 433–449. <https://doi.org/10.1080/0969594X.2011.582838>
- Tal, R. T., Dori, Y. J., & Lazarowitz, R. (2000). A project-based alternative assessment system. *Studies In Educational Evaluation*, 26(2), 171–191. [https://doi.org/10.1016/S0191-491X\(00\)00014-6](https://doi.org/10.1016/S0191-491X(00)00014-6)
- Taras, M. (2008). Summative and formative assessment: Perceptions and realities. *Active Learning in Higher Education*, 9(2), 172–192. <https://doi.org/10.1177/1469787408091655>
- TEQSA. (2023). Assessment reform for the age of artificial intelligence. <https://www.teqsa.gov.au/guides-resources/resources/corporate-publications/assessment-reform-age-artificial-intelligence>
- Thanh, B. N., Vo, D. T. H., Nhat, M. N., Pham, T. T. T., Trung, H. T., & Xuan, S. H. (2023). Race with the machines: Assessing the capability of generative AI in solving authentic assessments. *Australasian Journal of Educational Technology*, 39(5), 59–81. <https://doi.org/10.14742/ajet.8902>
- The University of Melbourne. (n.d.). Peer assisted study sessions. <https://arts.unimelb.edu.au/students/undergraduate/enrich-your-bachelor-of-arts/peer-assisted-study-sessions>
- The University of Sydney. (2023). What you need to know about the new academic integrity policy. <https://www.sydney.edu.au/study/student-life/student-news/2023/05/29/what-you-need-to-know-about-the-new-academic-integrity-policy.html>
- Tiruneh, D. T., Verburgh, A., & Elen, J. (2014). Effectiveness of critical thinking instruction in higher education: A systematic review of intervention studies. *Higher Education Studies*, 4(1), 1–17. <https://doi.org/10.5539/hes.v4n1p1>
- UNESCO. (2023). Guidance for generative AI in education and research. <https://www.unesco.org/en/articles/guidance-generative-ai-education-and-research>
- University of Bath. (2023). Support for students to use GenAI tools effectively and responsibly. <https://www.bath.ac.uk/announcements/support-for-students-to-use-genai-tools-effectively-and-responsibly/>
- University of Oxford. (n.d.). AI in teaching and assessment. <https://academic.admin.ox.ac.uk/ai-in-teaching-and-assessment>
- UTS. (n.d.). GenAI learning resources. <https://www.uts.edu.au/current-students/support/helps/self-help-resources/genai-learning-resources>
- Van den Bergh, V., Mortelmans, D., Spoor, P., Van Petegem, P., Gijbels, D., & Vanthournout, G. (2006). New assessment modes within project-based education-the stakeholders. *Studies In Educational Evaluation*, 32(4), 345–368. <https://doi.org/10.1016/j.stueduc.2006.10.005>
- Vuopala, E., Hyvönen, P., & Järvelä, S. (2016). Interaction forms in successful collaborative learning in virtual learning environments. *Active Learning in Higher Education*, 17(1), 25–38. <https://doi.org/10.1177/1469787415616730>
- Walters, K. S. (1990). Critical thinking, rationality, and the vulcanization of students. *The Journal of Higher Education*, 61(4), 448–467. <https://doi.org/10.1080/00221546.1990.11780724>
- Wang, Y., Liu, C., & Tu, Y. F. (2021). Factors affecting the adoption of AI-based applications in higher education. *Educational Technology & Society*, 24(3), 116–129.
- Wangaard, D. B., & Stephens, J. M. (2011). *Creating a culture of academic integrity: A toolkit for secondary schools*. Search Institute Press.
- Wood, P. (2023). Oxford and Cambridge ban AI language tool GPT-3 over fears of plagiarism. The i <https://inews.co.uk/news/oxford-cambridge-ban-chatgpt-plagiarism-universities-2178391>
- Yilmaz, R., & Yilmaz, F. G. K. (2023). The effect of generative artificial intelligence (AI)-based tool use on students' computational thinking skills, programming self-efficacy and motivation. *Computers in Education: Artificial Intelligence*, 4, Article 100147. <https://doi.org/10.1016/j.caeai.2023.100147>
- Zawacki-Richter, O., Marin, V. I., Bond, M., & Gouverneur, F. (2019). Systematic review of research on artificial intelligence applications in higher education—Where are the educators? *International Journal of Educational Technology in Higher Education*, 16(1), 1–27. <https://doi.org/10.1186/s41239-019-0171-0>